



Vishal S R 250701844-2025-CSE ▾

**V2****Started on** Friday, 7 August 2026, 9:19 AM**State** Finished**Completed on** Friday, 7 August 2026, 9:32 AM**Time taken** 13 mins 10 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include<stdio.h>
2
3  void function(int n)
4  {
5      int count=1;
6      int c= 0;count++;
7      for(int i=n/2; i<n; i++)
8      {
9          count+=2;
10         for(int j=1; j<n; j = 2 * j)
11         {
12             count+=2;
13             for(int k=1; k<n; k = k * 2)
14             {
15                 c++;
16                 count+=2;
17             }
18         }
19     }
20     printf("%d",count);
21 }
22 int main(){
23     int n;
24     scanf("%d",&n);
25     function(n);
26 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 30       | 30  | ✓ |
| ✓ | 10    | 212      | 212 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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